

Knowledge Graphs for Evidence-Based Rehabilitation

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First International Workshop on Knowledge Graphs for
Health (KG4Health)



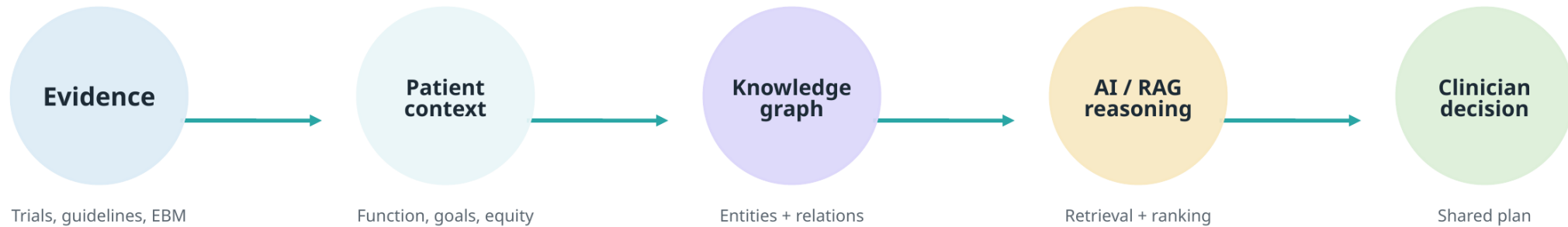
Why this topic matters now

- Rehabilitation is personalized, context-dependent, and longitudinal.
- Evidence is fragmented across trials, guidelines, clinical notes, biomechanics, and lived experience.
- Current AI often predicts or classifies, but does not always explain why or what to do next.
- Clinicians need systems that preserve meaning, evidence quality, and human oversight.

KGs can act as a semantic bridge between data, evidence, and decisions.

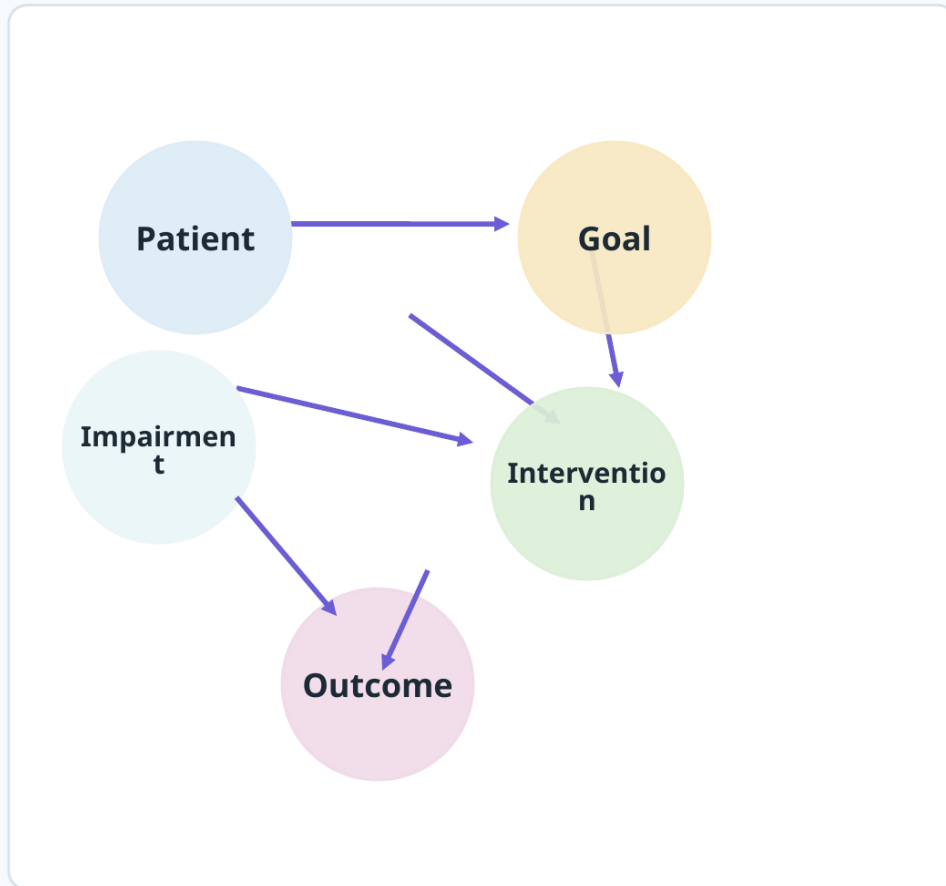


Mental model: from data to clinical reasoning



The aim is not to replace clinical reasoning; it is to make the evidence trail and assumptions visible.

What is a knowledge graph?



Nodes

clinical concepts, patient factors, interventions, outcomes, evidence, settings

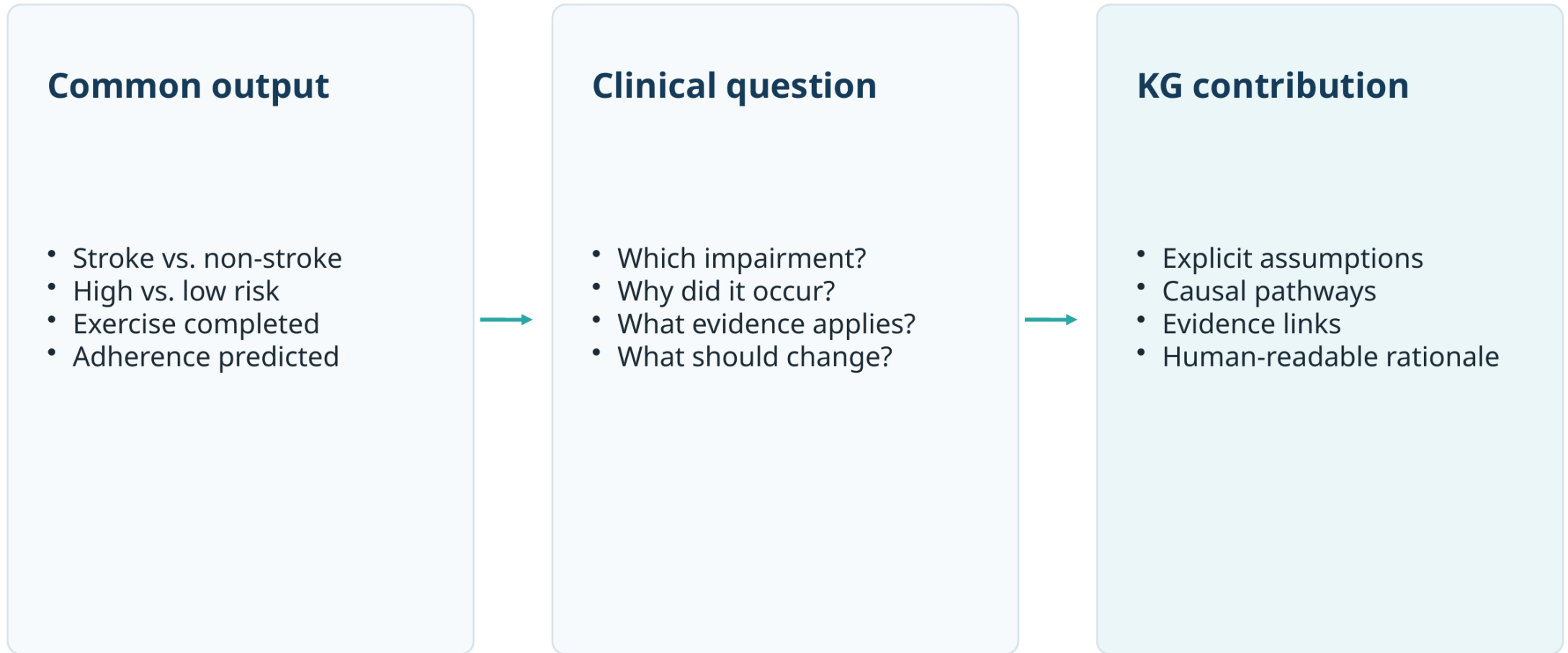
Edges

meaningful relationships such as causes, treats, contraindicates, measured by, or applies to

Value

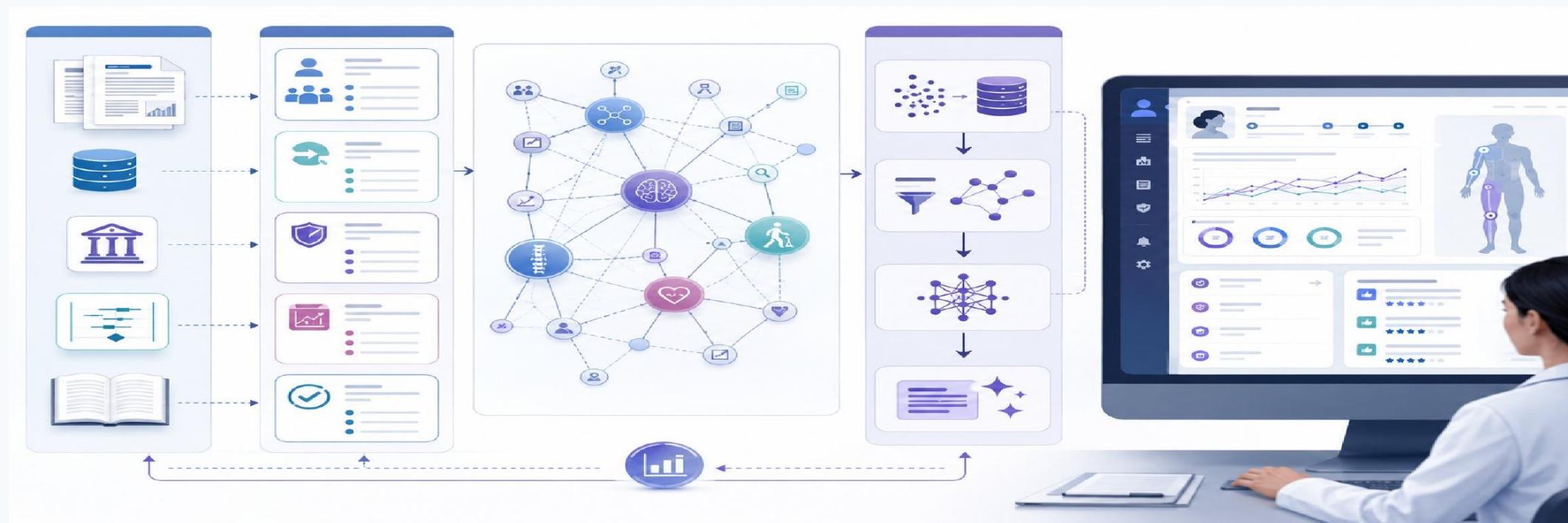
semantic integration, traceability, explanation, retrieval, and updating as evidence evolves

Rehabilitation AI has an interpretation gap



EVIDENCE-AWARE AI

KG + RAG can make rehabilitation AI more evidence-based



PICO/PICOT alignment

Evidence hierarchy

Faithful answers

Example 1: personalized rehabilitation design



Problem

Rehabilitation devices and VR tasks need to adapt to physiological signals, behaviour, preferences, and design parameters.

KG role

A rehabilitation product design KG can semantically integrate fNIRS, behavioural feedback, and product variables.

Takeaway

KGs can translate multi-source data into personalized design recommendations rather than isolated metrics.

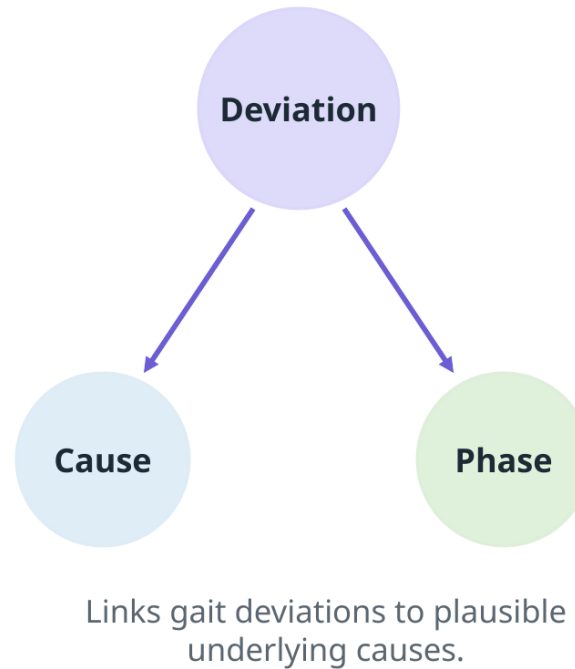
Case reference: Wang et al., Advanced Engineering Informatics, 2026

Example 2: post-stroke gait assessment

Inputs

- Kinematic gait data
- Normative reference data
- Clinical gait videos
- Expert stroke rehabilitation knowledge

GDC-EKG



Outputs

- Detected deviations
- Potential causes
- Clinician review
- Exportable reasoning trail

Pilot evidence: high expert agreement and favourable technology acceptance, but larger validation is needed.

Example 3: Case study: Knowledge graph embeddings for pain

Article used in this section

Chaturvedi, J., Wang, T., Velupillai, S., Stewart, R., & Roberts, A. (2024)
Development of a Knowledge Graph Embeddings Model for Pain
AMIA Annual Symposium Proceedings, 2023, 299–308

Pain is multidimensional and context-dependent.
Much pain information appears in free-text clinical notes.
SNOMED CT provides standardized clinical relationships.
Knowledge graph embeddings make these relationships usable
for prediction.

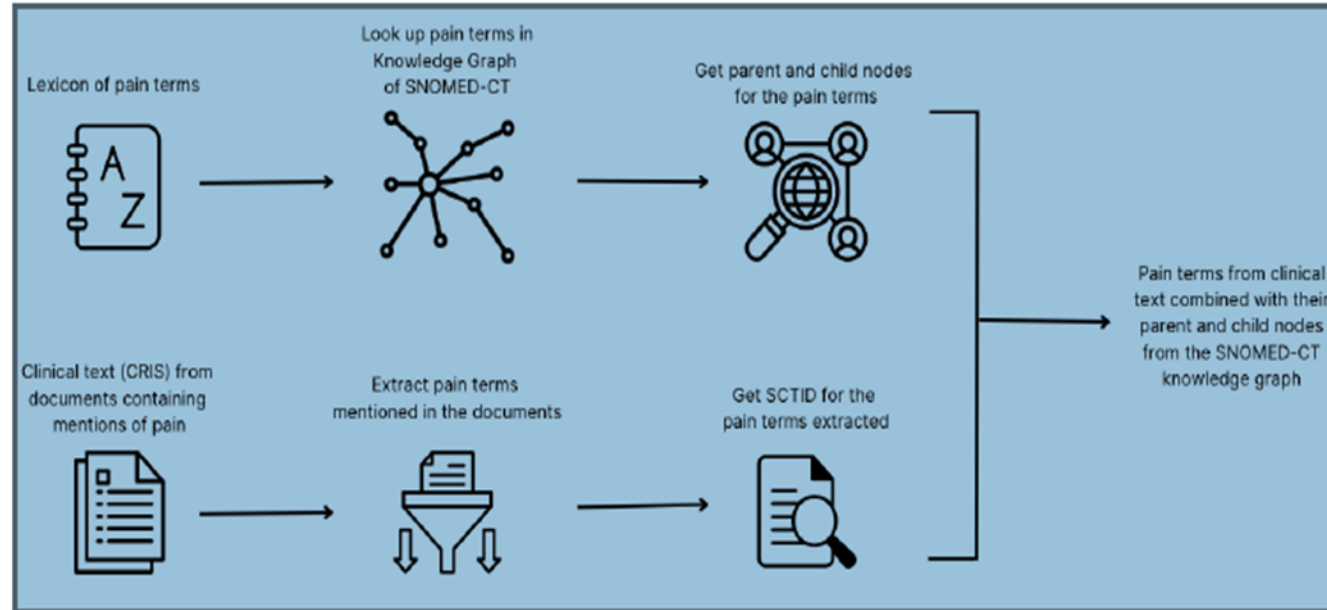


Figure 1. Creation of dataset for building KGE models

Why pain is a useful example for knowledge graphs



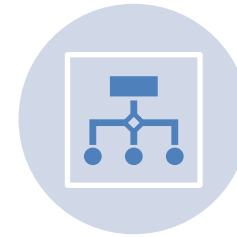
AA PAIN CAN BE LINKED TO
DISORDERS, INJURIES,
MEDICATIONS, SYMPTOMS,
AND FUNCTION.



PATIENTS AND CLINICIANS
DESCRIBE PAIN USING
DIFFERENT WORDS AND
LEVELS OF DETAIL.



IMPORTANT CONTEXT MAY BE
EMBEDDED IN NARRATIVE
NOTES.



A GRAPH CAN ORGANIZE
THESE CONCEPTS AND
RELATIONS IN A COMPUTABLE
FORM.

What the pain KGE study did

Started with pain-related terms

The authors used a pain lexicon to identify relevant pain keywords.

Searched clinical notes for pain mentions

They extracted free-text documents from electronic health records that contained pain-related language.

Mapped pain concepts to SNOMED CT

Pain mentions were linked to standardized clinical concepts using SNOMED CT identifiers.

Extracted graph relationships

Parent and child nodes related to the pain concepts were obtained from the SNOMED CT knowledge graph.

Combined clinical text with external knowledge

Pain concepts from EHR notes were combined with SNOMED CT relationships to build a pain knowledge graph.

Built knowledge graph embedding models

The graph was transformed into embeddings using ComplEx and TransE models.

Evaluated the models with link prediction

The models were tested on whether they could predict missing relationships between pain-related concepts.

What does a clinical relationship look like?

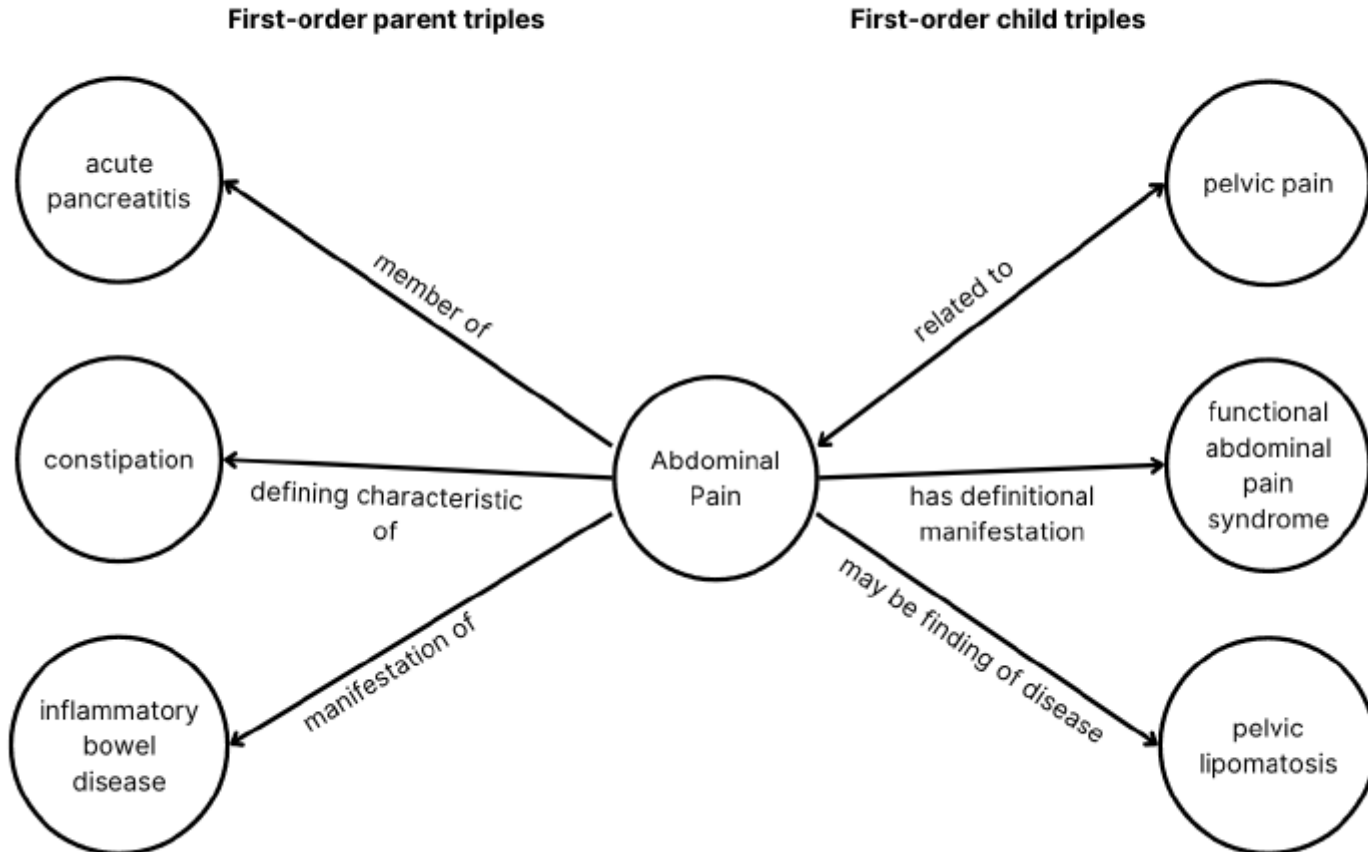


Figure 2. An example of first-order parent and child triples for the concept “abdominal pain”

Why knowledge graph embeddings matter



Knowledge graph embeddings provide a way to represent complex clinical relationships in a form that machine learning models can use.



Instead of relying only on isolated codes or words, the model learns from the structure of the graph: concepts, relations, and clinical context.



In the pain KGE study, the authors evaluated the embeddings through a link prediction task.



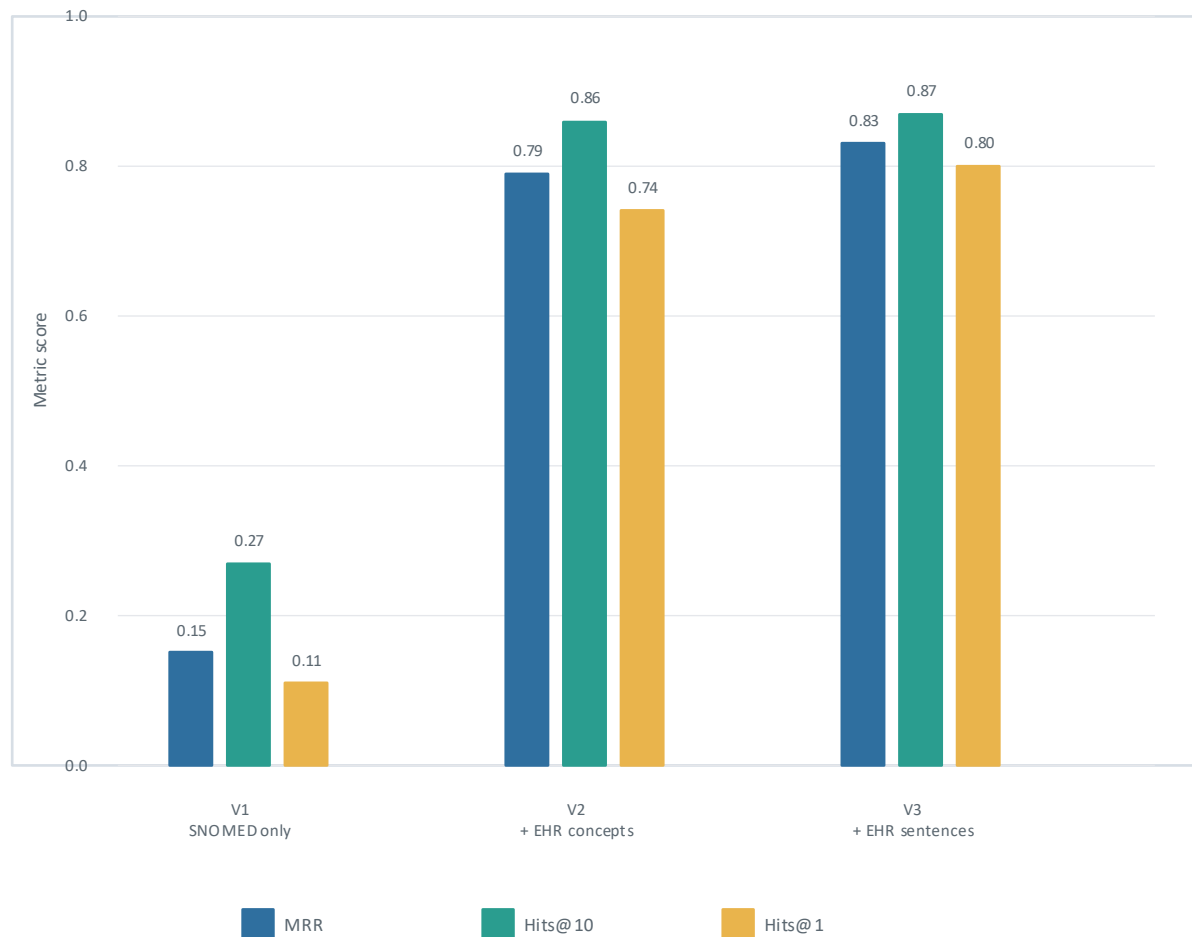
Link prediction tests whether the model can infer a missing subject or object in a known clinical relationship.



This helps assess whether the graph has captured clinically meaningful semantic patterns.

What the study found—and why it matters for rehabilitation

Complex model performance across three data variations



- Structured terminology alone was not enough.
- Adding real-world EHR concepts substantially improved performance.
- Sentence context provided the best Complex results.
- For rehabilitation, context is central: symptoms, function, setting, access, and care pathways.

What this enables in telerehabilitation



Remote assessment

Connect patient-reported outcomes, video-derived movement, sensors, and clinical notes.

Personalized recommendations

Map goals, impairments, contraindications, context, and evidence to possible next steps.

Equity-aware implementation

Represent access barriers, language, rurality, cultural safety, data governance, and community priorities.

A KG can help ask: “For whom does this evidence apply, under what conditions, and with what safeguards?”

Responsible implementation: four design requirements

1

Evidence traceability

Every recommendation should link back to sources, evidence quality, and inclusion/exclusion logic.

2

Clinical interpretability

Clinicians must see the relationships, assumptions, and uncertainty behind outputs.

3

Workflow fit

Outputs must be actionable within real rehabilitation workflows and time constraints.

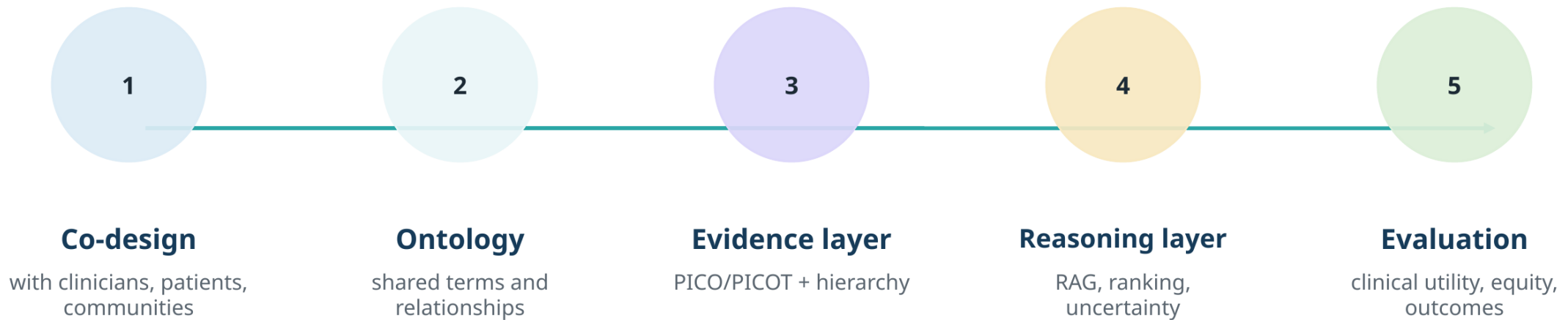
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Data governance

Community, patient, and institutional governance must shape what data are used and how.

Successful KG systems are socio-technical systems, not only technical architectures.

A research agenda for evidence-based rehabilitation KGs



Key evaluation question: does the system improve reasoning, decisions, access, and outcomes—not only model accuracy?

Three take-home messages

1

KGs preserve meaning

They connect patient context, clinical knowledge, and evidence in a machine-readable but human-reviewable structure.

2

RAG needs evidence logic

Retrieval should consider PICO/PICOT alignment, evidence hierarchy, uncertainty, and clinical fit.

3

Implementation is ethical work

Trustworthy systems require co-design, governance, equity, and clinician control.

Thank you

Questions and discussion

Selected references

- Wang, B., Qu, J., Li, X., Wang, C., & Bu, L. (2026). A personalized rehabilitation design method via knowledge graph-based multi-source data integration. *Advanced Engineering Informatics*.
- Jiao, Y., Liu, Z., Reading, S., Smith, M.-C., Lin, J., & Zhang, Y. (2025). A knowledge graph-based post-stroke gait assessment system: A pilot study. *Medical Engineering & Physics*.
- Zhang, J., et al. (2026). From Evidence-Based Medicine to Knowledge Graph: Retrieval-Augmented Generation for Sports Rehabilitation and a Domain Benchmark. *arXiv:2601.00216*.
- Hogan, A., et al. (2021). Knowledge graphs. *ACM Computing Surveys*.
- Ghassemi, M., Oakden-Rayner, L., & Beam, A. L. (2021). The false hope of current approaches to explainable AI in health care. *The Lancet Digital Health*.

Thank you

- AIRE Lab: <https://www.airereseach.ca/>
- Focus: equity-centered Digital Health and Rehabilitation

